Let us construct a sinusoidal function of the form $y = d + a\cos(bt - c)$ to find the depth of water at time t. Here, $a = 1.4$; $d = 2.8$; $p = 12$; $b = \frac{\pi}{6}$; $c = \frac{\pi}{3}$.

The required sinusoidal function is $y = 2.8 + 1.4\cos\left(\frac{\pi}{6}t - \frac{\pi}{3}\right)$.

Note

The transformations of sine and cosine functions are useful in numerous applications. A circular motion is always modelled using either the sine or cosine function.

Model-2

A point rotates around a circle with centre at origin and radius 4. We can obtain the y-coordinate of the point as a function of the angle of rotation.

For a point on a circle with centre at the origin and radius a, the y-coordinate of the point is $y = a\sin\theta$, where θ is the angle of rotation. In this case, we get the equation $y(\theta) = 4\sin\theta$, where θ is in radian, the amplitude is 4 and the period is 2π . The amplitude 4 causes a vertical stretch of the y-values of the function $\sin\theta$ by a factor of 4.

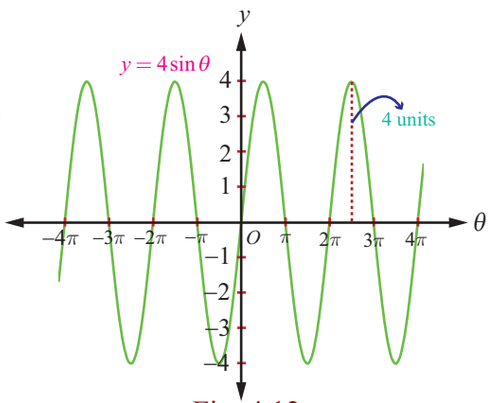


Fig. 4.12

4.4.3 The inverse cosine function and its properties

The cosine function is not one-to-one in the entire domain $\mathbb{R}$. However, the cosine function is one-to-one on the restricted domain $[0, \pi]$ and still, on this restricted domain, the range is $[-1, 1]$. Now, let us define the inverse cosine function with $[-1, 1]$ as its domain and with $[0, \pi]$ as its range.

Definition 4.4

For $-1 \le x \le 1$, define $\cos^{-1} x$ as the unique number y in $[0, \pi]$ such that $\cos y = x$. In other words, the inverse cosine function $\cos^{-1} : [-1, 1] \to [0, \pi]$ is defined by $\cos^{-1}(x) = y$ if and only if $\cos y = x$ and $y \in [0, \pi]$.

Note

(i) The sine function is non-negative on the interval $[0, \pi]$, the range of $\cos^{-1} x$. This observation is very important for some of the trigonometric substitutions in Integral Calculus.

(ii) Whenever we talk about the inverse cosine function, we have $\cos x : [0, \pi] \to [-1, 1]$ and $\cos^{-1} x : [-1, 1] \to [0, \pi]$.

(iii) We can also restrict the domain of the cosine function to any one of the intervals $\cdots, [-\pi, 0], [\pi, 2\pi], \cdots$, where it is one-to-one and its range is $[-1, 1]$.

The restricted domain $[0, \pi]$ is called the **principal domain** of cosine function and the values of $y = \cos^{-1} x, -1 \le x \le 1$, are known as **principal values** of the function $y = \cos^{-1} x$.

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